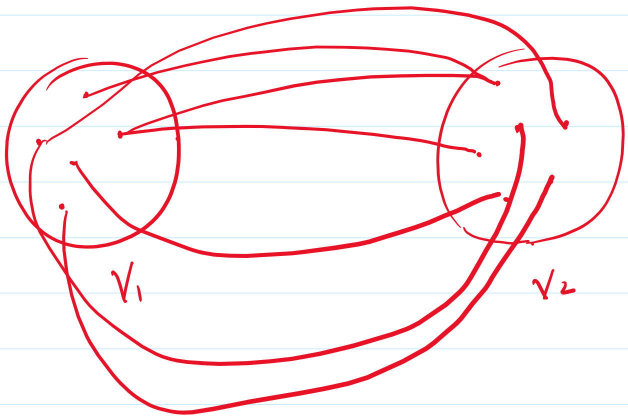
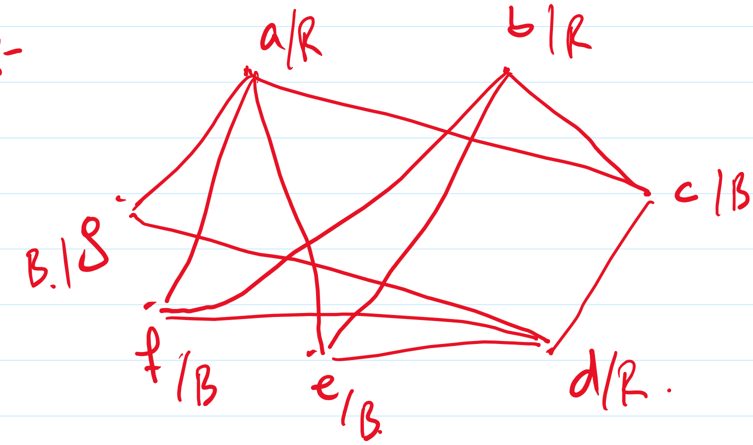
Lecture 18.

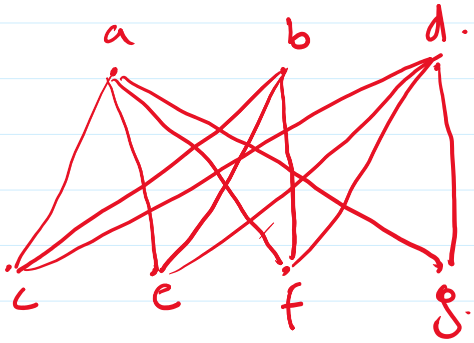
Bipartite.



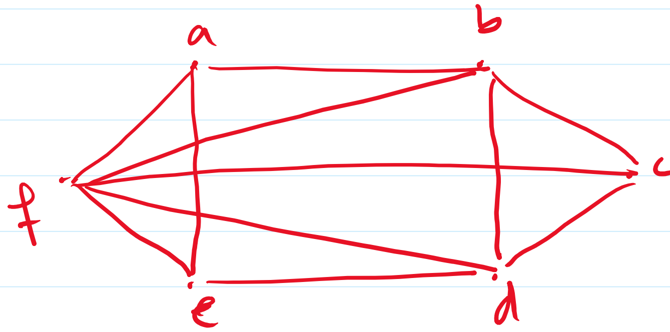
Ex11 : 541



Two adjacent must have different Colour.

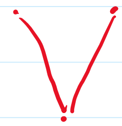
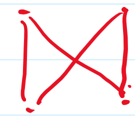


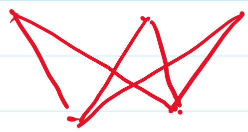
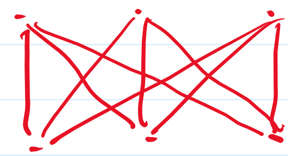
Ex11 541.



X.

5- Complete Bi-partite. $K_{m,n}$. $m \geq 1$ $n \geq 1$.

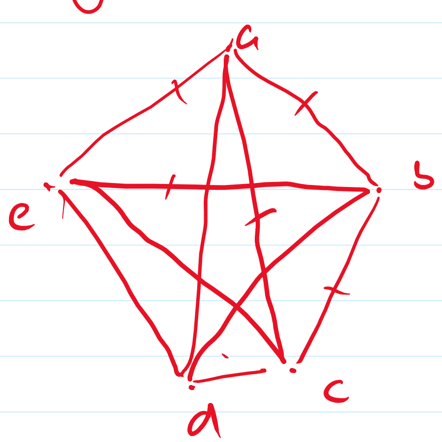
$K_{1,1}$  $K_{2,1}$  $K_{2,2}$ 

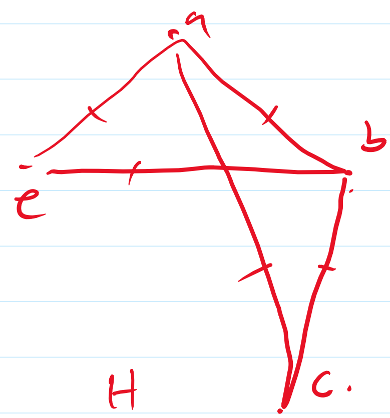
$K_{3,2}$  $K_{3,3}$ 

	Vertices	Edges.
$K_{1,1}$	2.	1.
$K_{2,1}$	3	2.
$K_{2,2}$	4	4.
$K_{3,2}$	5	6
$K_{3,3}$	6	9.
⋮	⋮	⋮
$K_{m,n}$	$m+n$	mn.

Subgraph. $G = (V, E)$ $H = (W, F)$.

H is a Subgraph of G.
if $W \subseteq V$ $\wedge$ $F \subseteq E$.



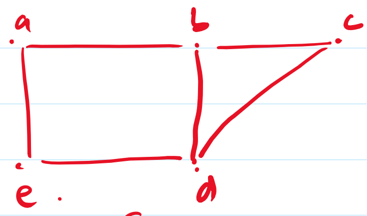


H

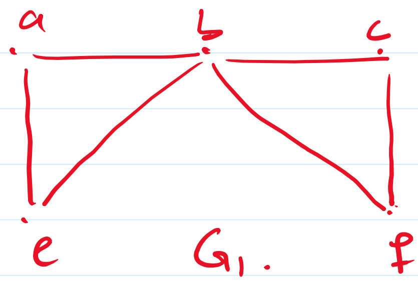
G .

P 546 .

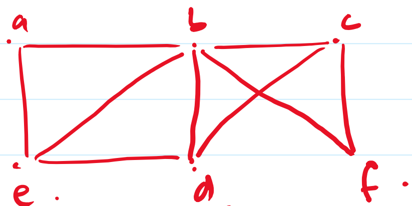
Ex 17



G_1 .



G_1 .



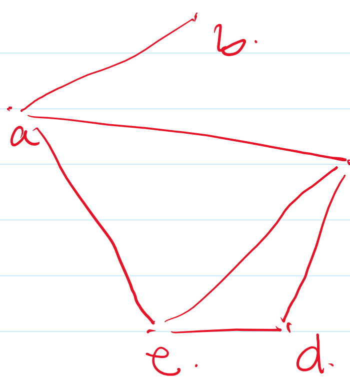
$G_1 \cup G_2$.

Representing Graphs .

1- Adjacency list .

Ex 1

P 550 .

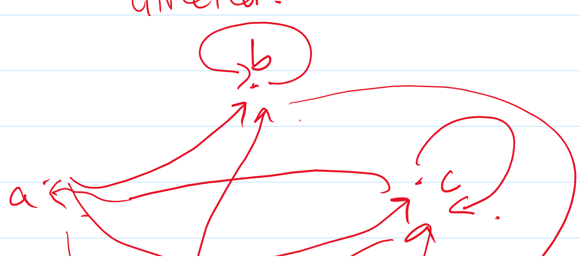


Vertex	adjacent Vertex .
a	e, c, b .
b	a
c	a, d, e .
d	c, e .
e .	d, a, c .

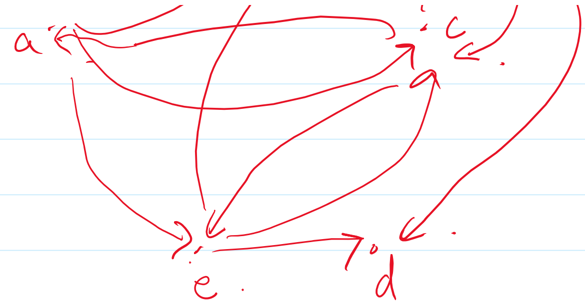
Ex 2

550 .

directed .



Initial Vertex	Terminal Vertex
a	b, c, e
b	b, d .
c	c, a, e .
d	—



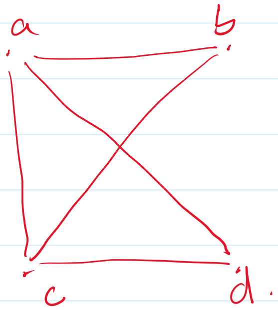
c, d, e.

d —

e . d, c, b.

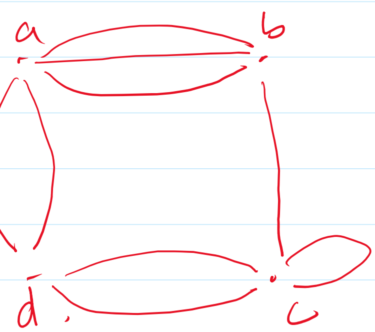
2- Adjacency Matrix.

Size = $|V| \times |V|$.



$$\begin{array}{c|cccc|c} & a & b & c & d & \\ \hline a & 0 & 1 & 1 & 1 & 3 \\ b & 1 & 0 & 1 & 0 & 2 \\ c & 1 & 1 & 0 & 1 & 3 \\ d & 1 & 0 & 1 & 0 & 2 \\ \hline & & & & & 10 \end{array}$$

Ex5
551.



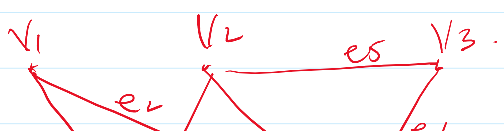
$$\begin{array}{c|cccc|} & a & b & c & d \\ \hline a & 0 & 3 & 0 & 2 \\ b & 3 & 0 & 1 & 0 \\ c & 0 & 1 & 1 & 2 \\ d & 2 & 0 & 2 & 0 \end{array}$$

Simple 1) Main diagonal Contains all 0's.
2). No entry > 1.

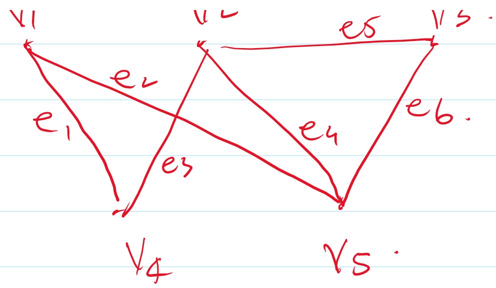
3-. Incidence Matrix

Size = $|V| \times |E|$.

Ex6
P552.



Ex6
P552.



	e_1	e_2	e_3	e_4	e_5	e_6
v_1	1	1	0	0	0	0
v_2	0	0	1	1	1	0
v_3	0	0	0	0	1	1
v_4	1	0	1	0	0	0
v_5	0	1	0	1	0	1

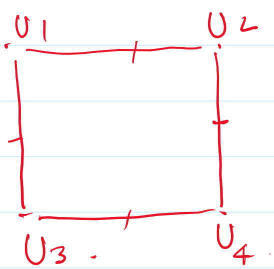
1 – Column wise Sum = 1
→ loop.

2 - Two Columns Same
→ multi edge.

Isomorphism.

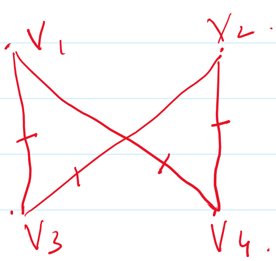
Two Simple Graphs $G_1 = (V_1, E_1)$ ∧ $G_2 = (V_2, E_2)$ are isomorphic if ∃ a one-to-one ∧ onto function f: from V_1 to V_2 such that if a, b are adjacent in G_1 then $f(a), f(b)$ are adjacent in G_2.

Ex8
553.



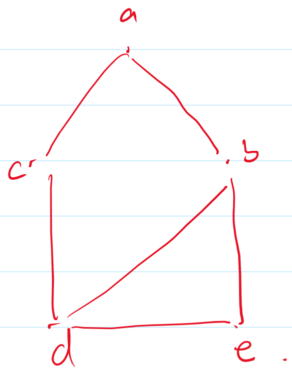
G_1

$f(u_1) = v_1$
$f(u_2) = v_4$
$f(u_3) = v_3$
$f(u_4) = v_2$

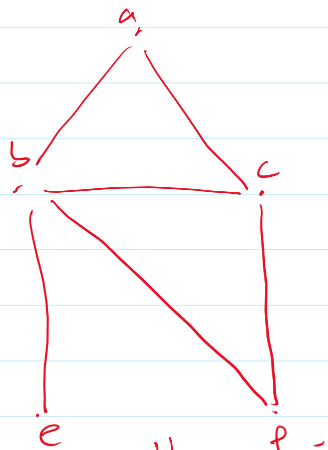


G_2

(u_1, u_2)	$f(u_1), f(u_2)$. v_1, v_4.	
(u_2, u_4)	$f(u_2), f(u_4)$	v_4, v_2.
(u_4, u_3)	$f(u_4), f(u_3)$.	v_2, v_3.
(u_3, u_1).	$f(u_3), f(u_1)$.	v_3, v_1



G.

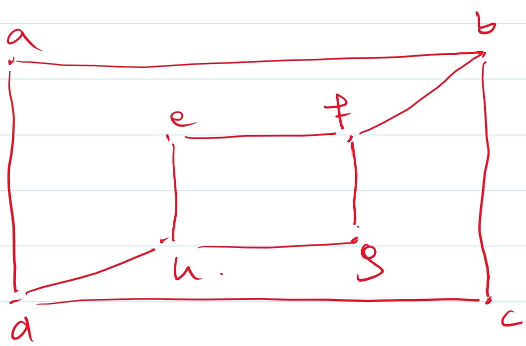


H.

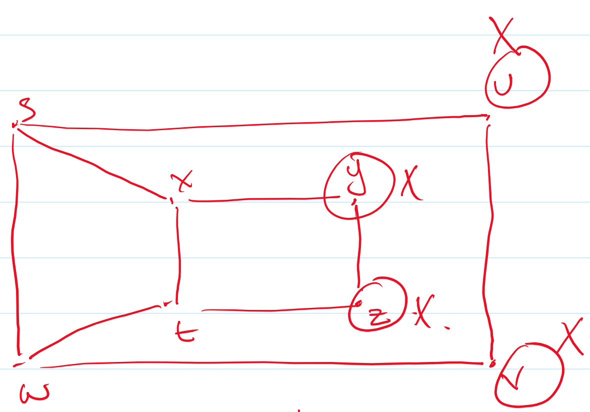
	G	H.
Vertex	5	5
Edges.	6	6
degrees	Does not Contain Vertex with degree 1.	Contain Vertex of degree 1.

X.

P554.



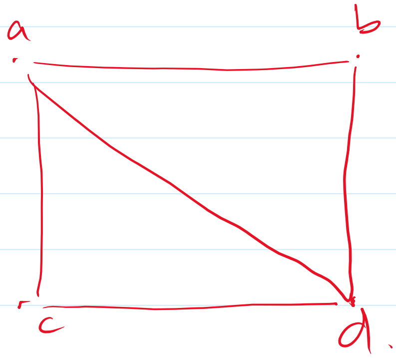
G

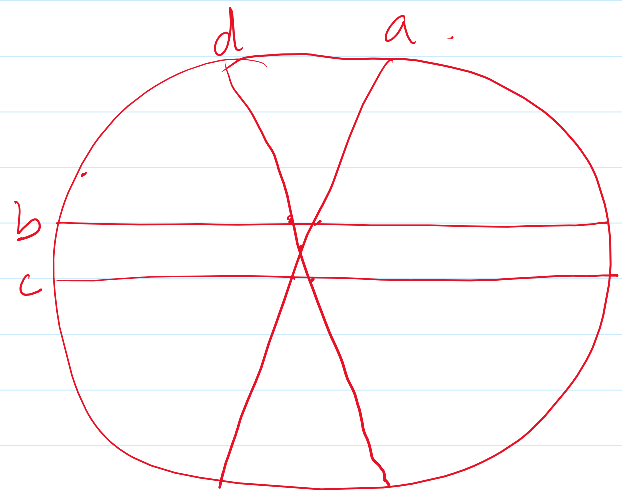


H.

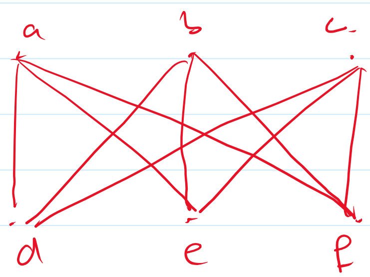
		G	H
1)	Vertex	8	8
2)	Edges	10	10.
3).	degree 2=4. 4 3=4		deg 2=4 deg 3=4.
4)	Adjacent degrees.		

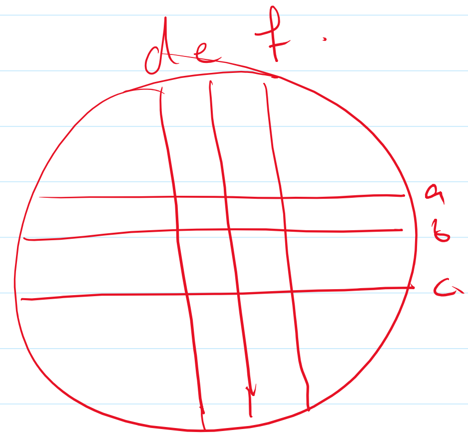
Circular Graph.



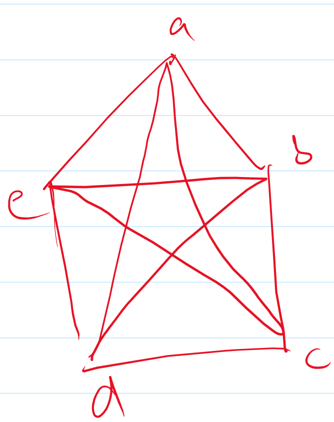


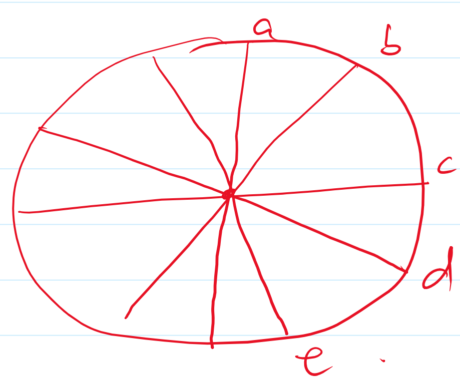
$K_{3,3}$.





K_5





W_3 Circular Graph.